📚 Advanced Java Programming — Comprehensive Study Notes

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Unit 1: Introduction to Java, Inheritance, Exception Handling, Multithreading, Applet Programming, Connecting to a Server,

Implementing Servers, Making URL Connections, Socket Programming

Topic 1.1: Introduction to Java

Overview

Java is a high-level, class-based, object-oriented programming language developed by Sun Microsystems (now owned by Oracle Corporation). It is platform-independent, meaning Java programs can run on any device with the Java Virtual Machine (JVM) installed. Java was designed to have as few implementation dependencies as possible with the "Write Once, Run Anywhere" (WORA) principle.

Key Definitions

> ***Java Virtual Machine (JVM):*** *A virtual machine that enables Java bytecode to be executed on any platform. It acts as an abstract computer that runs Java bytecode.*

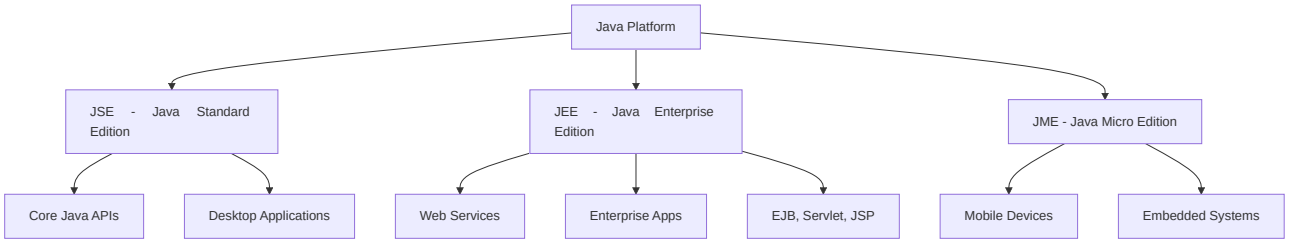
> ***Bytecode:*** *Platform-independent compiled Java code with .class extension. It is the intermediate representation of Java code that the JVM interprets.*

> ***WORA (Write Once, Run Anywhere):*** *The principle that Java code written on one platform can run on any other platform that has a JVM without recompilation.*

Java Key Features

Feature	Description
Platform Independent	Works on "write once, run anywhere" principle. Once you write Java code, it can run on any platform with Java installed.
Object-Oriented	Everything in Java is an object. Supports OOP concepts: Object, Class, Inheritance, Polymorphism, Abstraction, Encapsulation.
Robust	Strong memory management system. Helps eliminate errors through compile-time and runtime checking.
Secure	Does not use explicit pointers and runs programs inside a sandbox to prevent untrusted activities.
Multi-threaded	Supports concurrent execution of multiple threads for maximum CPU utilization.
Portable	Java programs can be easily moved from one computer system to another.

Java Editions



- **JSE (Java Standard Edition):** Core Java for standalone desktop applications
- **JEE (Java Enterprise Edition):** Advanced Java for enterprise-level distributed applications, web services, and server-side development
- **JME (Java Micro Edition):** For mobile devices and embedded systems

Basic Syntax Rules

```java
public class MyFirstJavaProgram {
    /* This is my first java program.
     * This will print 'Hello World' as the output
     */
    public static void main(String []args) {
        System.out.println("Hello World");
    }
}
```

Syntax Rules:

- **Case Sensitivity:** Java is case sensitive - `Hello` and `hello` are different identifiers
- **Class Names:** First letter should be in Upper Case (e.g., `MyClass`)
- **Method Names:** Start with lower case, inner words capitalized (e.g., `myMethod`)
- **File Name:** Must match class name exactly
- **Main Method:** `public static void main(String args[])` is mandatory for application entry point

Java Identifiers

All Java components require names. Rules:

- Begin with a letter (A-Z, a-z), currency character ($), or underscore (_)
- After first character, can have any combination
- Keywords cannot be used as identifiers

Legal identifiers: `age` , `$salary` , `_value` , `__1_value` **Illegal identifiers:** `123abc` , `-salary`

Java Modifiers

Access Modifiers: `default` , `public` , `protected` , `private`

Non-access Modifiers: `final` , `abstract` , `strictfp`

Variables

Type	Description
Local Variables	Declared inside methods, blocks, or constructors
Instance Variables (Non-static)	Declared in a class, unique to each object
Class Variables (Static)	Shared across all instances of a class

Key Takeaways

- ✅ Java provides platform independence through JVM
- ✅ JSE is the foundation; JEE extends it for enterprise applications
- ✅ JME targets mobile and embedded devices
- ✅ Java uses bytecode for platform independence

Cross-References

- Related to: Topic 1.5 (Applet Programming), Topic 1.9 (Socket Programming)

Topic 1.2: Inheritance

Overview

Inheritance is a fundamental concept in object-oriented programming (OOP) that allows a new class to inherit properties and behavior from an existing class. In Java, inheritance enables code reuse and establishes relationships between classes through a parent-child hierarchy.

Key Definitions

> ***Inheritance:*** *A mechanism where a class acquires the properties and behaviors of another class, promoting code reuse and establishing "is-a" relationships.*

> ***Superclass (Parent Class):*** *The class whose properties and methods are inherited by another class.*

> ***Subclass (Child Class):*** *The class that inherits properties and methods from another class.*

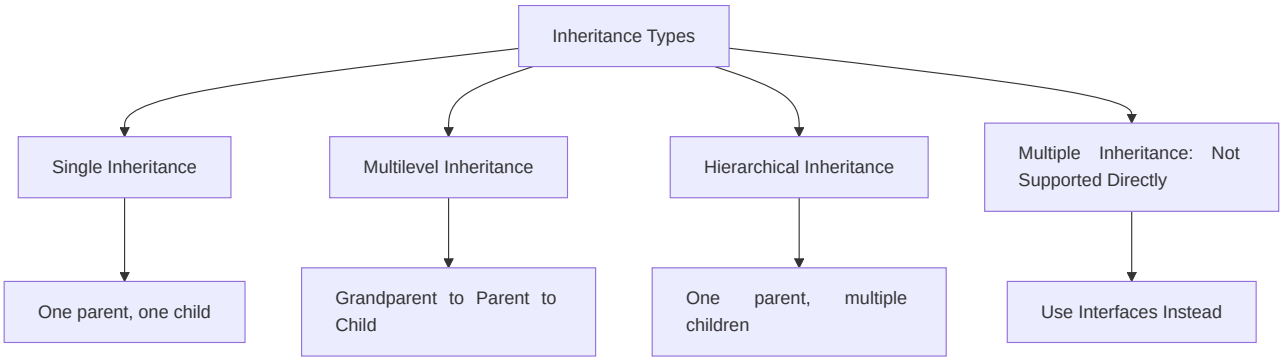
> ***Polymorphism:*** *The ability of objects to take on many forms; allows writing code that deals with general object types.*

Syntax

```
class Superclass {
    // superclass members
}

class Subclass extends Superclass {
    // subclass members
}
```

Types of Inheritance



1. Single Inheritance

In single inheritance, a subclass inherits from only one superclass.

```
class Animal { }
class Dog extends Animal { }
```

2. Multilevel Inheritance

In multilevel inheritance, a subclass inherits from a superclass, and another class inherits from this subclass.

```
class Animal { }
class Mammal extends Animal { }
class Dog extends Mammal { }
```

3. Hierarchical Inheritance

In hierarchical inheritance, multiple subclasses inherit from a single superclass.

```
class Animal { }
class Dog extends Animal { }
class Cat extends Animal { }
```

4. Multiple Inheritance (Not Supported)

Java does not support multiple inheritance of classes due to the "diamond problem." Use interfaces instead:

```
interface A { void methodA(); }
interface B { void methodB(); }

class CombinedClass implements A, B {
    public void methodA() { System.out.println("Method A"); }
    public void methodB() { System.out.println("Method B"); }
}
```

Example of Inheritance

```java
// Superclass
class Vehicle {
    void move() {
        System.out.println("Vehicle is moving");
    }
}

// Subclass inheriting from Vehicle
class Car extends Vehicle {
    void accelerate() {
        System.out.println("Car is accelerating");
    }
}

// Main class
public class Main {
    public static void main(String[] args) {
        Car car = new Car();
        car.move();        // Output: Vehicle is moving
        car.accelerate();  // Output: Car is accelerating
    }
}
```

Key Points

- **Method Overriding:** Subclasses can provide a specific implementation for a method defined in the superclass
- **Access Modifiers:** Inherited members can have different visibility (public, protected, private)
- **Constructor Inheritance:** Constructors are not inherited, but subclass constructor implicitly calls `super()`
- **final Keyword:** Methods marked final cannot be overridden

Key Takeaways

- ✅ Inheritance enables "is-a" relationship
- ✅ Composition uses "has-a" relationship
- ✅ Single inheritance only in Java (use interfaces for multiple)
- ✅ Use `super()` to access parent class methods and constructors

Cross-References

- Related to: Topic 2.1 (JavaBean), Topic 2.2 (Session Beans)

Topic 1.3: Exception Handling

Overview

Exception handling is a mechanism provided by Java to handle runtime errors, known as exceptions, in a structured and graceful manner. By handling exceptions effectively, developers can prevent unexpected program terminations and provide meaningful feedback to users.

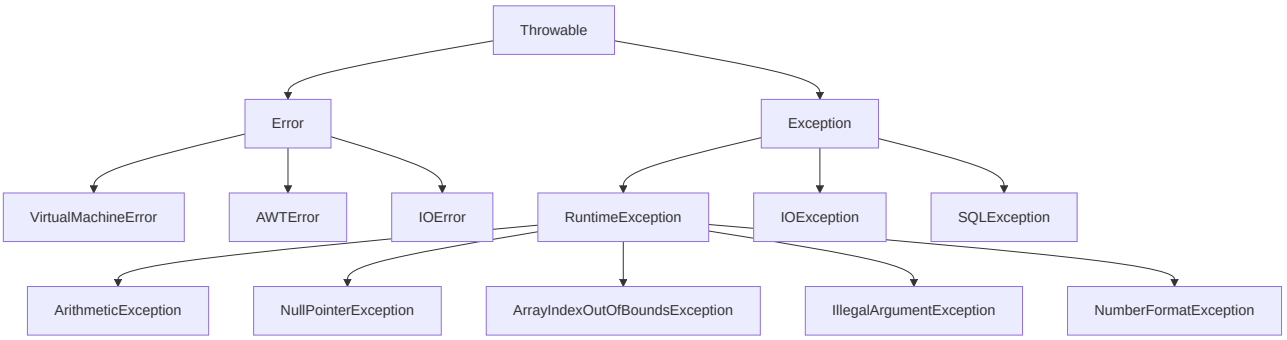
Key Definitions

> ***Exception:*** *An event that disrupts the normal flow of the program's execution. It is an unwanted event that occurs during program execution.*

> ***Throwable:*** *The root class of the exception hierarchy in Java. All exceptions inherit from this class.*

> ***try-catch:*** *A block of code where exceptions are handled. The try block contains code that might throw an exception, and catch blocks handle specific exception types.*

Exception Hierarchy



Types of Exceptions

Type	Description	Examples
Checked Exceptions	Compiled-time exceptions that must be handled or declared	IOException, SQLException, ClassNotFoundException
Unchecked Exceptions	Runtime exceptions not checked at compile time	ArithmeticException, NullPointerException, ArrayIndexOutOfBoundsException

Exception Handling Constructs

1. **try:** Block surrounding program statements to monitor for exceptions
2. **catch:** Catches specific kinds of exceptions and handles them
3. **finally:** Code that must execute whether or not an exception occurs
4. **throw:** Used to throw a specific exception from the program
5. **throws:** Specifies which exceptions a method can throw

Syntax

```
try {
    // Code that may throw an exception
} catch (ExceptionType1 e1) {
    // Handle ExceptionType1
} catch (ExceptionType2 e2) {
    // Handle ExceptionType2
} finally {
    // Code that always executes, regardless of whether an exception occurred or not
}
```

Code Examples

Example 1: Arithmetic Exception

```java
public class Main {
    public static void main(String[] args) {
        try {
            int result = 10 / 0; // ArithmeticException: / by zero
        } catch (ArithmeticException e) {
            System.out.println("Error: " + e.getMessage());
        }
    }
}
```

Example 2: ArrayIndexOutOfBoundsException

```java
public class Main {
    public static void main(String[] args) {
        int[] array = {1, 2, 3};
        try {
            int value = array[3]; // ArrayIndexOutOfBoundsException
        } catch (ArrayIndexOutOfBoundsException e) {
            System.out.println("Error: " + e.getMessage());
        }
    }
}
```

Example 3: NullPointerException

```java
public class Main {
    public static void main(String[] args) {
        String str = null;
        try {
            int length = str.length(); // NullPointerException
        } catch (NullPointerException e) {
            System.out.println("Error: " + e.getMessage());
        }
    }
}
```

Example 4: Handling Multiple Exceptions

```java
public class Main {
    public static void main(String[] args) {
        try {
            int[] array = {1, 2, 3};
            int value = array[3]; // ArrayIndexOutOfBoundsException
            int result = 10 / 0;   // ArithmeticException
        } catch (ArrayIndexOutOfBoundsException e) {
            System.out.println("Array Index Error: " + e.getMessage());
        } catch (ArithmeticException e) {
            System.out.println("Arithmetic Error: " + e.getMessage());
        }
    }
}
```

Example 5: throw and throws

```java
public class NumberUtils {
    // throws declares potential exceptions
    public static int divide(int numerator, int denominator) throws ArithmeticException {
        if (denominator == 0) {
            // throw explicitly throws an exception
            throw new ArithmeticException("Division by zero!");
        }
        return numerator / denominator;
    }

    public static void main(String[] args) {
        try {
            int result = divide(10, 0);
            System.out.println("Result: " + result);
        } catch (ArithmeticException e) {
            System.out.println("Error: " + e.getMessage());
        }
    }
}
```

throw vs throws

throw	throws
Used within method to throw exception	Declared in method signature
Takes exception object as argument	Informs caller about potential exceptions
Triggers immediate termination	Does not throw exception itself

Key Takeaways

- ✅ Checked exceptions must be handled or declared with throws
- ✅ Unchecked exceptions occur at runtime
- ✅ Multiple catch blocks must order from specific to general
- ✅ finally block always executes (except System.exit())
- ✅ Custom exceptions extend Exception class

Cross-References

- Related to: Topic 1.4 (Multithreading), Topic 2.4 (Servlet Exception Handling)

Topic 1.4: Multithreading

Overview

Multithreading is a programming concept that allows multiple threads to execute concurrently within a single process. In Java, multithreading enables developers to create applications that can perform multiple tasks simultaneously, improving efficiency and responsiveness.

Key Definitions

> ***Thread:*** *A lightweight subprocess, the smallest unit of processing that can be scheduled by the operating system. A thread is a single sequential flow of control within a program.*

> ***Process:*** *An independent program execution that has its own memory space. A process can contain multiple threads.*

> ***Runnable Interface:*** *An interface that provides a way to create a thread without extending the Thread class.*

Thread vs Process

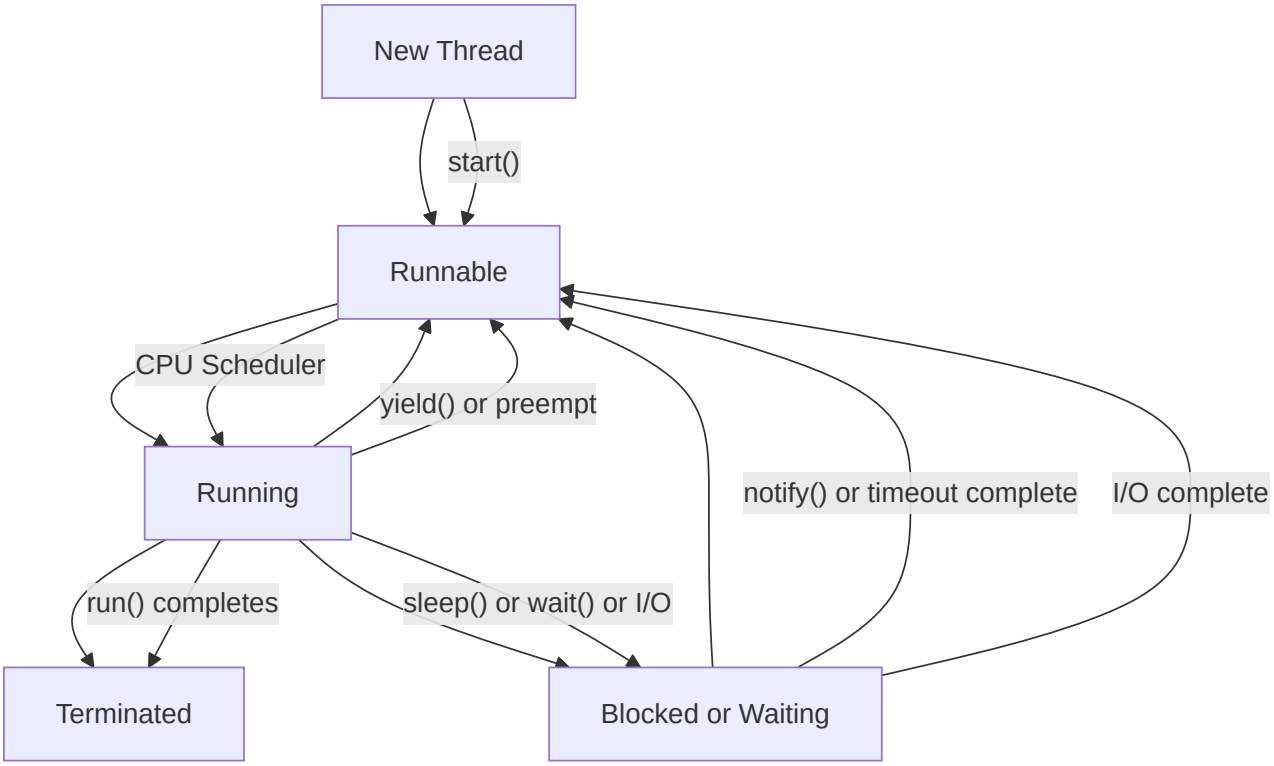
Characteristic	Process	Thread
Memory	Separate memory space	Shares memory with other threads
Creation	Heavyweight, slower	Lightweight, faster
Communication	IPC required	Direct communication possible
Independence	Independent	Dependent on process

Thread Lifecycle States

A thread can exist in the following states:

1. **New:** Thread object created but not started
2. **Runnable:** Thread is ready to run, waiting for CPU
3. **Running:** Thread is executing
4. **Blocked/Waiting:** Thread is waiting for resources or another thread
5. **Terminated:** Thread has completed execution

Visual: Thread Lifecycle



Creating Threads in Java

There are two primary ways to create threads in Java:

1. **By extending Thread class**
2. **By implementing Runnable interface**

Code Examples

Method 1: Extending Thread Class

```java
class MyThread extends Thread {
    public void run() {
        for (int i = 1; i <= 5; i++) {
            System.out.println("Thread: " + i);
            try {
                Thread.sleep(1000); // Pause for 1 second
            } catch (InterruptedException e) {
                System.out.println(e);
            }
        }
    }
}

public class Main {
    public static void main(String[] args) {
        MyThread thread = new MyThread();
        thread.start(); // Start the thread
    }
}
```

Method 2: Implementing Runnable Interface

```java
class MyRunnable implements Runnable {
    public void run() {
        for (int i = 1; i <= 5; i++) {
            System.out.println("Runnable: " + i);
            try {
                Thread.sleep(1000); // Pause for 1 second
            } catch (InterruptedException e) {
                System.out.println(e);
            }
        }
    }
}

public class Main {
    public static void main(String[] args) {
        Thread thread = new Thread(new MyRunnable());
        thread.start(); // Start the thread
    }
}
```

Method 3: Thread Synchronization

```java
class Counter {
    private int count = 0;

    public synchronized void increment() {
        count++;
    }

    public int getCount() {
        return count;
    }
}

class MyThread extends Thread {
    private Counter counter;

    public MyThread(Counter counter) {
        this.counter = counter;
    }

    public void run() {
        for (int i = 0; i < 1000; i++) {
            counter.increment();
        }
    }
}

public class Main {
    public static void main(String[] args) throws InterruptedException {
        Counter counter = new Counter();
        MyThread thread1 = new MyThread(counter);
        MyThread thread2 = new MyThread(counter);

        thread1.start();
        thread2.start();

        thread1.join();
        thread2.join();

        System.out.println("Count: " + counter.getCount());
    }
}
```

Thread Pool Example (Image Downloader)

```java
import java.util.concurrent.ExecutorService;
import java.util.concurrent.Executors;

public class ImageDownloader {
    public static void main(String[] args) {
        String[] imageUrls = {"url1", "url2", "url3", "url4", "url5"};
        ExecutorService executor = Executors.newFixedThreadPool(imageUrls.length);

        for (String url : imageUrls) {
            executor.execute(new ImageDownloadTask(url));
        }

        executor.shutdown();
    }
}

class ImageDownloadTask implements Runnable {
    private String imageUrl;

    public ImageDownloadTask(String imageUrl) {
        this.imageUrl = imageUrl;
    }

    @Override
    public void run() {
        try {
            // Download logic here
            System.out.println("Downloaded: " + imageUrl);
        } catch (Exception e) {
            e.printStackTrace();
        }
    }
}
```

Thread Lifecycle Methods Summary

Method	Description
start()	Causes the thread to begin execution
run()	Contains the code to be executed in the thread
sleep(long ms)	Pauses thread execution for specified milliseconds
join()	Waits for thread to complete execution
wait()	Causes thread to wait until notified (from Object class)
notify()	Wakes up a single waiting thread
notifyAll()	Wakes up all waiting threads
yield()	Causes thread to temporarily pause and allow other threads
interrupt()	Interrupts a sleeping or waiting thread

Key Takeaways

- ✅ Threads share memory space within a process, enabling efficient communication
- ✅ Multiple threads can exist within the same process
- ✅ Runnable interface is preferred over extending Thread (allows class to extend other class)
- ✅ Synchronized keyword prevents race conditions in shared resources
- ✅ Thread lifecycle: New → Runnable → Running → Blocked → Terminated

Cross-References

- Related to: Topic 1.3 (Exception Handling), Topic 1.7 (Implementing Servers)

Topic 1.5: Applet Programming

Overview

Applet programming in Java enables developers to create dynamic and interactive content that can be embedded within web pages. Java applets provide a platform-independent way to enhance web pages with features such as animations, games, and interactive forms.

Key Definitions

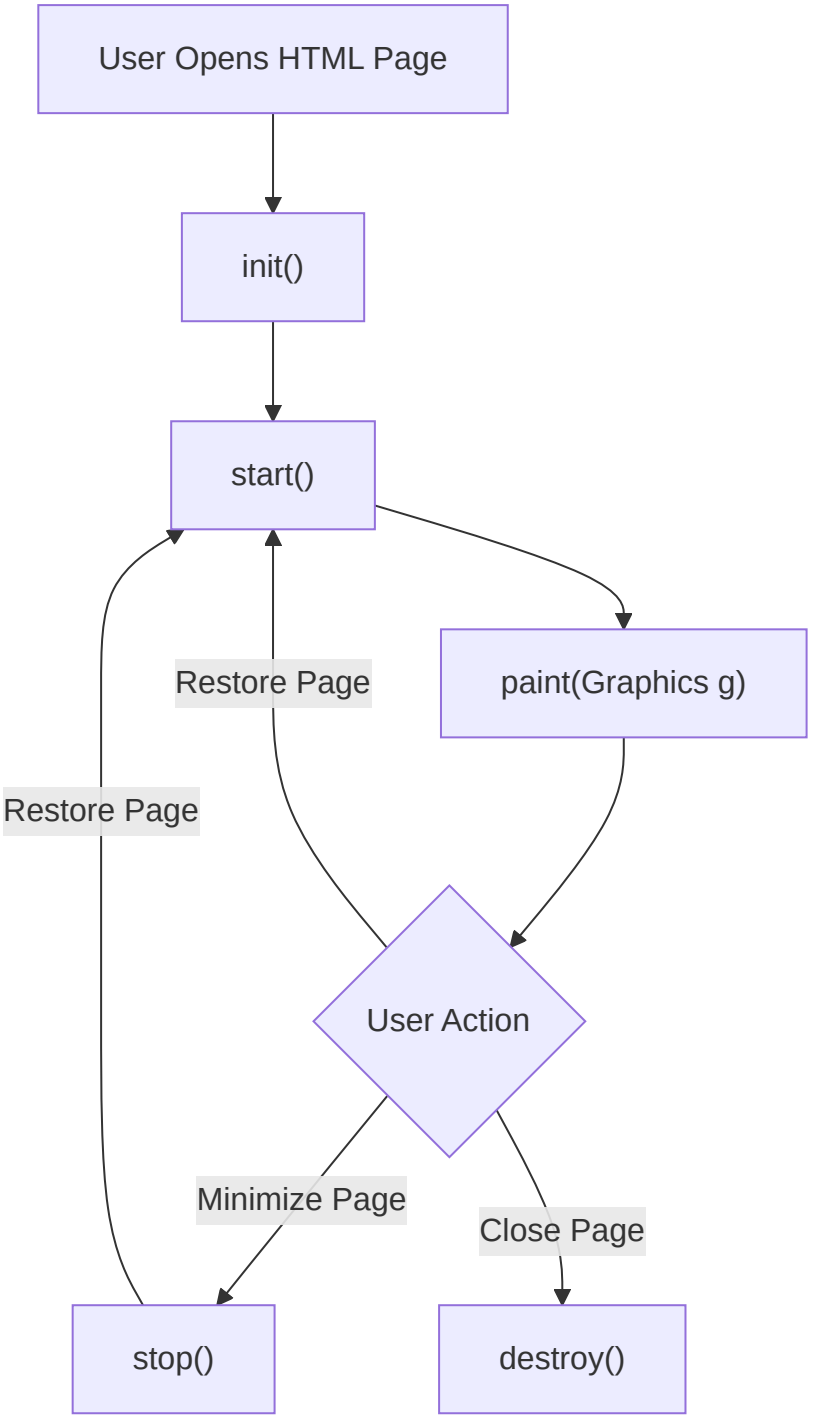
> ***Applet:*** *A Java program that runs within a web browser. It is a special Java program embedded in an HTML page.*

> ***Applet Lifecycle:*** *The sequence of methods called from initialization to destruction of an applet.*

Applet Lifecycle Methods

Method	Description
init()	Called once when applet is first loaded
start()	Called after init() and whenever applet restarts
paint(Graphics g)	Called when applet needs to be redrawn
stop()	Called when applet is stopped (e.g., minimized)
destroy()	Called when applet is terminated

Visual: Applet Lifecycle



Code Examples

Simple Applet

```java
import java.applet.Applet;
import java.awt.Graphics;

public class HelloWorldApplet extends Applet {
    public void paint(Graphics g) {
        g.drawString("Hello, World!", 20, 20);
    }
}
```

Drawing Shapes

```java
import java.applet.Applet;
import java.awt.Graphics;

public class ShapeApplet extends Applet {
    public void paint(Graphics g) {
        g.drawRect(20, 20, 100, 50);    // Draw a rectangle
        g.drawOval(150, 20, 80, 80);    // Draw an oval
        g.drawLine(300, 20, 400, 100);   // Draw a line
    }
}
```

Mouse Events

```java
import java.applet.Applet;
import java.awt.Graphics;
import java.awt.event.MouseEvent;
import java.awt.event.MouseListener;

public class MouseApplet extends Applet implements MouseListener {
    int x = 0;
    int y = 0;

    public void init() {
        addMouseListener(this); // Register mouse listener
    }

    public void paint(Graphics g) {
        g.drawString("Click at (" + x + ", " + y + ")", 20, 20);
    }

    // MouseListener methods
    public void mouseClicked(MouseEvent e) {
        x = e.getX(); // Get X-coordinate of mouse click
        y = e.getY(); // Get Y-coordinate of mouse click
        repaint(); // Refresh applet
    }

    public void mousePressed(MouseEvent e) {}
    public void mouseReleased(MouseEvent e) {}
    public void mouseEntered(MouseEvent e) {}
    public void mouseExited(MouseEvent e) {}
}
```

Animation

```java
import java.applet.Applet;
import java.awt.Graphics;

public class AnimationApplet extends Applet implements Runnable {
    int x = 0;

    public void init() {
        Thread t = new Thread(this);
        t.start(); // Start the animation thread
    }

    public void paint(Graphics g) {
        g.drawString("Moving Text", x, 20);
    }

    public void run() {
        while (true) {
            x += 5; // Move text horizontally
            if (x > getWidth()) {
                x = 0; // Reset position
            }
            repaint(); // Refresh applet
            try {
                Thread.sleep(100); // Pause for 100 milliseconds
            } catch (InterruptedException e) {
                System.out.println(e);
            }
        }
    }
}
```

HTML APPLET Tag Attributes

Attribute	Description
CODE	Name of the applet class file (required)
WIDTH	Width of applet display area (required)
HEIGHT	Height of applet display area (required)
CODEBASE	Base URL of applet code
ALT	Alternative text for browsers
NAME	Name for applet instance
ALIGN	Alignment (LEFT, RIGHT, TOP, etc.)
VSPACE/HSPACE	Vertical/horizontal spacing

Applet Advantages

- ✅ Works on client-side, reduces server load
- ✅ Secure - runs in sandbox
- ✅ Platform-independent

Applet Drawbacks

- ❌ Requires Java plugin in browser
- ❌ Limited functionality due to security restrictions
- ❌ Deprecated in modern browsers

Key Takeaways

- ✅ Applets extend Applet or JApplet class
- ✅ No main() method - lifecycle managed by browser/viewer
- ✅ Graphics output via paint() method

- ✅ Security restrictions apply (no file access, network restrictions)

Cross-References

- Related to: Topic 1.6 (Connecting to Server), Topic 1.9 (Socket Programming)

Topic 1.6: Connecting to a Server

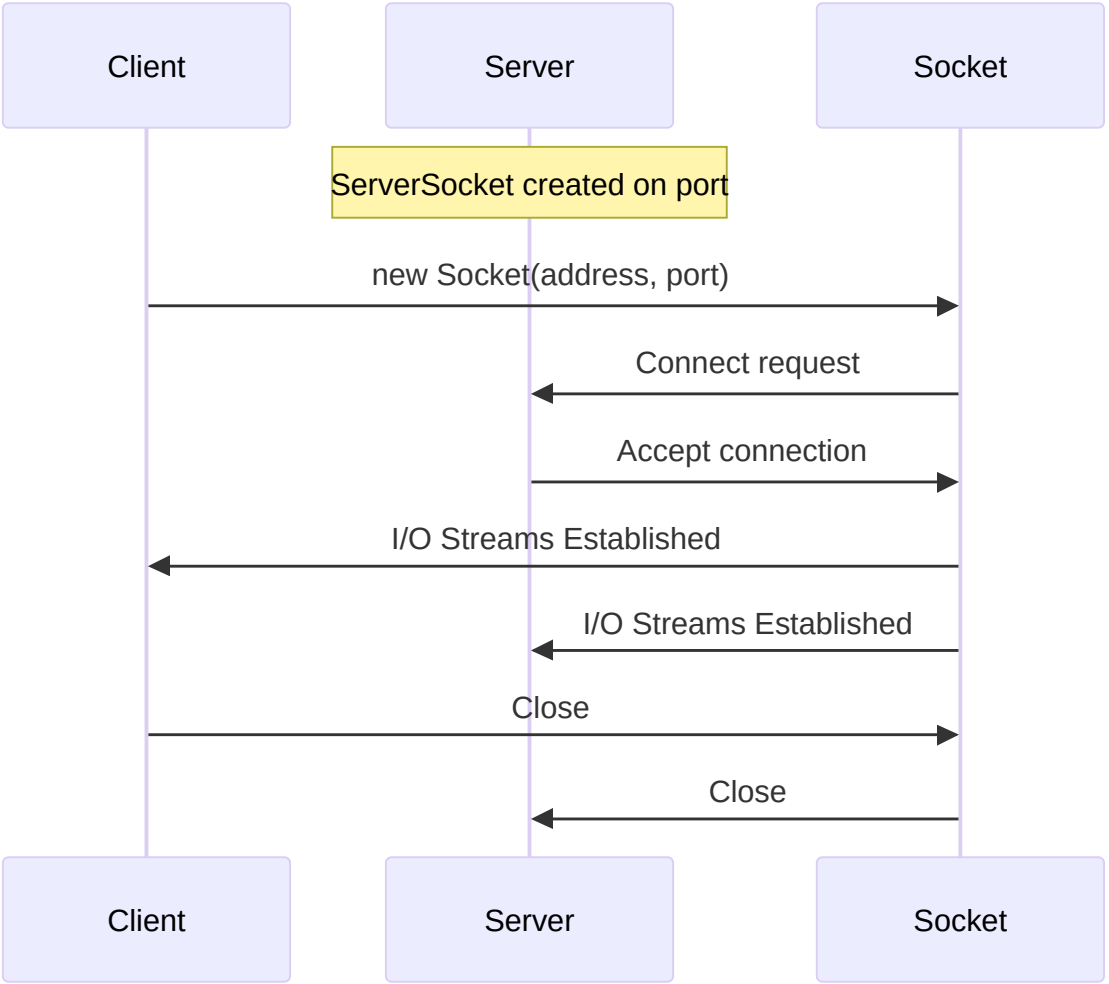
Overview

Connecting to a server in Java typically involves network communication, where a client application communicates with a server over a network protocol such as TCP/IP. Java provides the Socket class for this purpose.

Steps to Connect to Server

1. **Create a Socket:** Specify server address and port number
2. **Get Input/Output Streams:** For reading and writing data
3. **Communicate:** Exchange data with the server
4. **Close Connection:** Clean up resources

Visual: Client-Server Connection



Code Examples

Client Code

```java
import java.io.*;
import java.net.*;

public class Client {
    public static void main(String[] args) {
        try {
            // Create a socket to connect to the server
            Socket socket = new Socket("server_address", port_number);

            // Get the output stream from the socket
            OutputStream outputStream = socket.getOutputStream();

            // Create a PrintWriter for writing to the output stream
            PrintWriter out = new PrintWriter(outputStream, true);

            // Send data to the server
            out.println("Hello, Server!");

            // Close the socket
            socket.close();
        } catch (IOException e) {
            e.printStackTrace();
        }
    }
}
```

Server Code

```java
import java.io.*;
import java.net.*;

public class Server {
    public static void main(String[] args) {
        try {
            // Create a server socket bound to a specific port
            ServerSocket serverSocket = new ServerSocket(port_number);

            // Listen for incoming connections from clients
            Socket clientSocket = serverSocket.accept();

            // Get the input stream from the client socket
            InputStream inputStream = clientSocket.getInputStream();

            // Create a BufferedReader for reading from the input stream
            BufferedReader in = new BufferedReader(new InputStreamReader(inputStream));

            // Read data from the client
            String message = in.readLine();
            System.out.println("Message from client: " + message);

            // Close the sockets
            clientSocket.close();
            serverSocket.close();
        } catch (IOException e) {
            e.printStackTrace();
        }
    }
}
```

Key Takeaways

- ✅ Socket combines IP address and port number for network communication
- ✅ ServerSocket listens on a specific port for incoming connections
- ✅ accept() blocks until client connects
- ✅ Handle IOException appropriately

Cross-References

- Related to: Topic 1.7 (Implementing Servers), Topic 1.9 (Socket Programming)

Topic 1.7: Implementing Servers

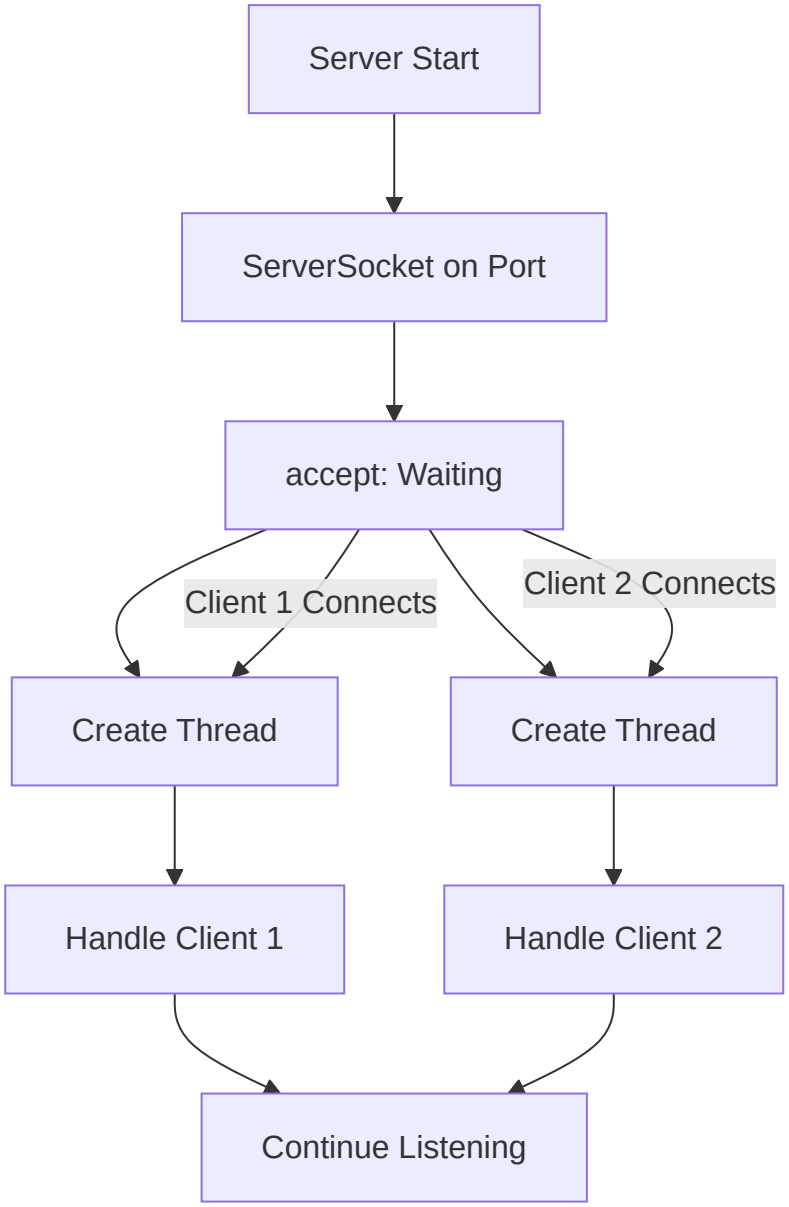
Overview

Implementing servers in Java involves creating applications that listen for incoming connections from clients and handle those connections according to the desired functionality. Java provides ServerSocket class for this purpose.

Server Implementation Steps

1. Create a ServerSocket on a specific port
2. Call accept() to wait for client connections
3. When a client connects, get the Socket object
4. Create input/output streams for communication
5. Process client requests
6. Close connections when done

Visual: Multi-Threaded Server



Code Examples

Server with Client Handler Thread

```java
import java.io.*;
import java.net.*;

public class Server {
    public static void main(String[] args) {
        try {
            // Create ServerSocket bound to port 12345
            ServerSocket serverSocket = new ServerSocket(12345);
            System.out.println("Server started. Waiting for clients...");

            // Accept client connections
            while (true) {
                Socket clientSocket = serverSocket.accept();
                System.out.println("Client connected: " + clientSocket);

                // Handle client connection in a separate thread
                ClientHandler clientHandler = new ClientHandler(clientSocket);
                clientHandler.start();
            }
        } catch (IOException e) {
            e.printStackTrace();
        }
    }
}

class ClientHandler extends Thread {
    private Socket clientSocket;

    public ClientHandler(Socket clientSocket) {
        this.clientSocket = clientSocket;
    }

    public void run() {
        try {
            // Get input and output streams from the client socket
            BufferedReader in = new BufferedReader(
                new InputStreamReader(clientSocket.getInputStream()));
            PrintWriter out = new PrintWriter(
                clientSocket.getOutputStream(), true);

            // Read data from the client
            String message = in.readLine();
            System.out.println("Message from client: " + message);

            // Send a response to the client
            out.println("Server received your message: " + message);

            // Close the client socket
            clientSocket.close();
        } catch (IOException e) {
            e.printStackTrace();
        }
```

```
    }
}
```

Additional Considerations

- **Thread Safety:** Ensure thread safety when accessing shared resources
- **Error Handling:** Handle exceptions appropriately, including closing resources
- **Scalability:** Consider scalability requirements and design the server to handle multiple concurrent clients
- **Protocol Design:** Define a protocol for communication between clients and server

Key Takeaways

- ✅ ServerSocket listens on a specific port
- ✅ accept() blocks until client connects
- ✅ Each client gets its own Socket
- ✅ Use threads for multiple concurrent clients

Cross-References

- Related to: Topic 1.9 (Socket Programming), Topic 2.4 (Servlet as Server-side)

Topic 1.8: Making URL Connections

Overview

Java provides classes for making URL connections and retrieving web resources. The URLConnection class represents a connection to a URL, while HttpURLConnection provides HTTP-specific functionality.

Key Definitions

> ***URL (Uniform Resource Locator):*** *A reference to a web resource that specifies its location on a computer network.*

> ***URLConnection:*** *An abstract class that represents a connection to a URL. It provides methods for reading header fields and managing the connection.*

URL Structure

```
protocol://hostname:port/path?query#fragment
```

URLConnection Methods

Method	Description
getInputStream()	Returns InputStream to read from server
getOutputStream()	Returns OutputStream to write to server
getContentEncoding()	Returns content encoding type
getContentLength()	Returns content length
getContentType()	Returns content type
getHeaderField()	Returns header field value

Code Examples

```java
import java.net.*;
import java.io.*;

public class URLReader {
    public static void main(String[] args) {
        try {
            // Define the URL
            URL url = new URL("https://example.com");

            // Open connection
            URLConnection connection = url.openConnection();

            // Get response code
            HttpURLConnection httpConnection = (HttpURLConnection) connection;
            int responseCode = httpConnection.getResponseCode();
            System.out.println("Response Code: " + responseCode);

            // Read content
            BufferedReader reader = new BufferedReader(
                new InputStreamReader(connection.getInputStream()));

            String line;
            while ((line = reader.readLine()) != null) {
                System.out.println(line);
            }
            reader.close();
        } catch (Exception e) {
            e.printStackTrace();
        }
    }
}
```

Key Takeaways

- ✅ URLConnection is abstract; use HttpURLConnection for HTTP
- ✅ getInputStream() for reading response
- ✅ getOutputStream() for sending data (POST)
- ✅ Header fields provide metadata about response

Cross-References

- Related to: Topic 1.6 (Connecting to Server), Topic 2.4 (Servlet Request/Response)

Topic 1.9: Socket Programming

Overview

Socket programming enables communication between applications running on different machines over a network. A socket is a combination of IP address and port number that identifies a specific endpoint for communication.

Key Definitions

> ***Socket:*** *A combination of IP address and port number used for network communication. It is the endpoint of a bidirectional communication link.*

> ***ServerSocket:*** *A socket that listens for incoming client connections on a specific port.*

Visual: Socket Connection Process

Key Classes

Class	Purpose
Socket	Client-side endpoint for connection
ServerSocket	Server-side endpoint for listening
InputStream	Read data from network
OutputStream	Write data to network

Code Examples

Simple Client

```java
import java.net.*;
import java.io.*;

public class Client {
    public static void main(String[] args) {
        try {
            Socket socket = new Socket("localhost", 5000);

            OutputStream outputStream = socket.getOutputStream();
            PrintWriter out = new PrintWriter(outputStream, true);

            out.println("Hello, Server!");

            socket.close();
        } catch (IOException e) {
            e.printStackTrace();
        }
    }
}
```

Simple Server

```java
import java.net.*;
import java.io.*;

public class Server {
    public static void main(String[] args) {
        try {
            ServerSocket serverSocket = new ServerSocket(5000);
            System.out.println("Server started...");

            Socket clientSocket = serverSocket.accept();
            System.out.println("Client connected");

            BufferedReader in = new BufferedReader(
                new InputStreamReader(clientSocket.getInputStream()));

            String message = in.readLine();
            System.out.println("Message: " + message);

            clientSocket.close();
            serverSocket.close();
        } catch (IOException e) {
            e.printStackTrace();
        }
    }
}
```

Key Takeaways

- ✅ Socket combines IP address and port number
- ✅ Port numbers range from 0 to 65535
- ✅ ServerSocket.accept() blocks until client connects
- ✅ Always close resources in finally block

Cross-References

- Related to: Topic 1.7 (Implementing Servers), Topic 4.2 (RMI)

Unit 2: JavaBean, Stateless Session Bean, Stateful Session Bean, Entity Bean, Servlet Overview and Architecture, Interface Servlet and the Servlet Life Cycle

Topic 2.1: JavaBean

Overview

JavaBeans are reusable software components that can be manipulated visually in a builder tool. A JavaBean is a Java class that follows specific conventions to enable visual manipulation and reuse.

Key Definitions

> ***JavaBean:*** *A reusable software component that follows specific conventions: no-arg constructor, Serializable, and property access via getter/setter methods.*

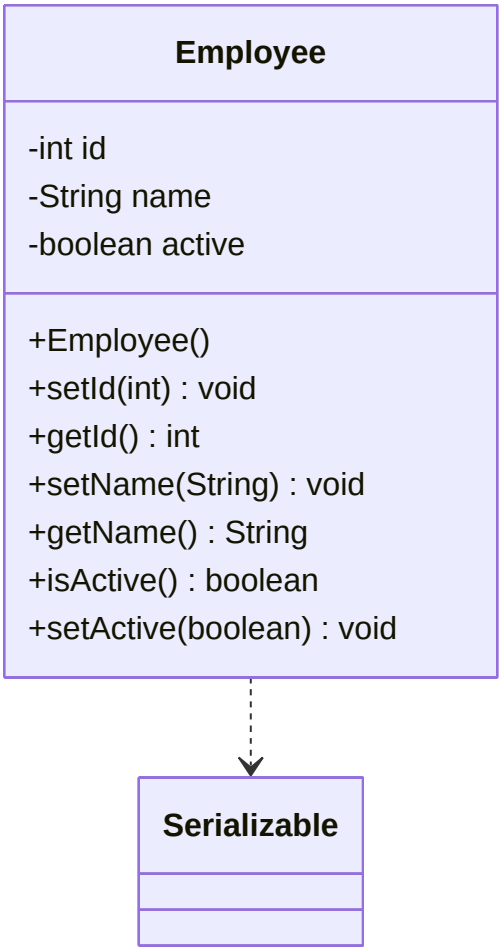
> ***Property:*** *A named feature of a JavaBean that can be accessed by users of the object through getter and setter methods.*

Requirements for a Java Bean

1. **Public no-arg constructor:** Must have a no-argument constructor

2. **Serializable:** Should implement `Serializable` interface
3. **Getters and Setters:** Properties accessed via `getXxx()` and `setXxx()` methods
4. **Naming convention:** Boolean properties can use `isXxx()` instead of `getXxx()`

Visual: JavaBean Structure



Code Example

```java
import java.io.Serializable;

public class Student implements Serializable {
    private static final long serialVersionUID = 1L;

    private String name;
    private int age;
    private String grade;

    // No-arg constructor
    public Student() { }

    // Parameterized constructor
    public Student(String name, int age, String grade) {
        this.name = name;
        this.age = age;
        this.grade = grade;
    }

    // Getters and Setters
    public String getName() { return name; }
    public void setName(String name) { this.name = name; }

    public int getAge() { return age; }
    public void setAge(int age) { this.age = age; }

    public String getGrade() { return grade; }
    public void setGrade(String grade) { this.grade = grade; }
}
```

Key Takeaways

- ✅ Follow the three conventions strictly (no-arg constructor, Serializable, getters/setters)
- ✅ Use meaningful property names
- ✅ For boolean properties, use "is" prefix for getters
- ✅ JavaBeans are fundamental to EJB and JSP

Cross-References

- Related to: Topic 2.2 (Session Beans), Topic 2.4 (Servlet)

Topic 2.2: Java Bean Properties

Overview

Java Bean properties define the state of a JavaBean. They can be accessed and modified through getter and setter methods.

Property Types

Type	Getter	Setter	Example
Read-Write	`getXxx()`	`setXxx()`	name
Read-Only	`getXxx()`	None	displayName (computed)
Write-Only	None	`setXxx()`	password
Boolean	`isXxx()`	`setXxx()`	active

Code Examples

```java
public class PropertyExample {
    private String name;
    private int age;
    private boolean active;
    private String password;

    // Read-Write property
    public String getName() { return name; }
    public void setName(String name) { this.name = name; }

    // Read-Only property (computed)
    public String getDisplayName() {
        return name + " (" + age + ")";
    }

    // Boolean property
    public boolean isActive() { return active; }
    public void setActive(boolean active) { this.active = active; }

    // Write-Only property
    public void setPassword(String password) {
        this.password = hash(password);
    }
}
```

Key Takeaways

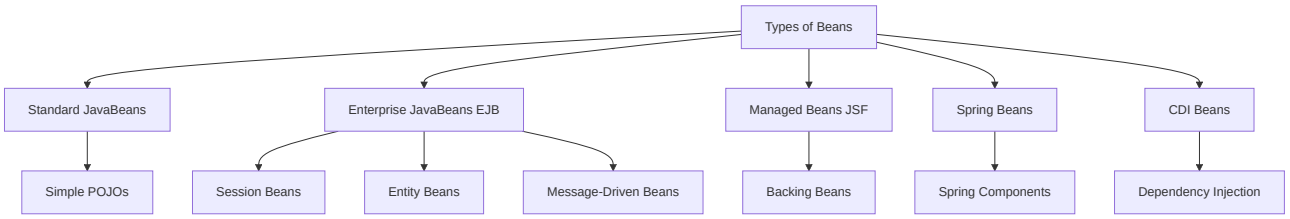
- ✅ Properties expose internal state through controlled access
- ✅ Encapsulation is maintained through getter/setter methods
- ✅ Read-only properties don't have setters
- ✅ Boolean properties use "is" prefix

Topic 2.3: Types of Beans

Overview

There are various types of beans in Java enterprise applications, each serving different purposes.

Types of Beans



Type	Description
Standard JavaBeans	Simple POJOs with properties
Enterprise JavaBeans (EJB)	Server-side business components
Managed Beans (JSF)	Backing beans for JSF pages
Spring Beans	Managed by Spring framework
CDI Beans	Contexts and Dependency Injection

Key Takeaways

- ✅ Different types of beans serve different architectural purposes

- ✅ Enterprise beans provide enterprise-level services
- ✅ Modern applications often use Spring or CDI

Topic 2.4: Stateful Session Bean

Overview

A Stateful Session Bean maintains conversational state across method calls within a session. It is dedicated to a single client for the duration of a session.

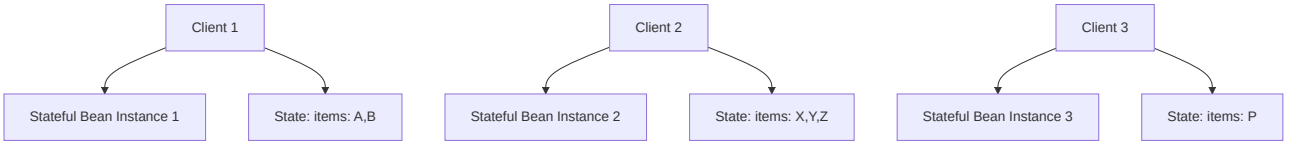
Key Definitions

> ***Stateful Session Bean:*** *A session bean that maintains conversational state with the client across multiple method calls.*

Characteristics

- ✅ Maintains client state across multiple requests
- ✅ One bean instance per client
- ✅ Removed when session ends
- ✅ Not persistent
- ✅ Cannot be pooled

Visual: Stateful Session Bean



Code Example

```java
import javax.ejb.Stateful;

@Stateful
public class ShoppingCartBean implements ShoppingCart {
    private List<String> items = new ArrayList<>();
    private String customerId;

    public void setCustomerId(String customerId) {
        this.customerId = customerId;
    }

    public void addItem(String item) {
        items.add(item);
    }

    public void removeItem(String item) {
        items.remove(item);
    }

    public List<String> getItems() {
        return items;
    }

    public void clear() {
        items.clear();
    }
}
```

Key Takeaways

- ✅ Maintains conversational state
- ✅ One-to-one relationship with client
- ✅ Not pooled like stateless beans
- ✅ Removed when session times out or client removes it

Topic 2.5: Stateless Session Bean

Overview

A Stateless Session Bean does not maintain conversational state with any client. Each bean instance can be used by any client, and its state is not preserved between method invocations.

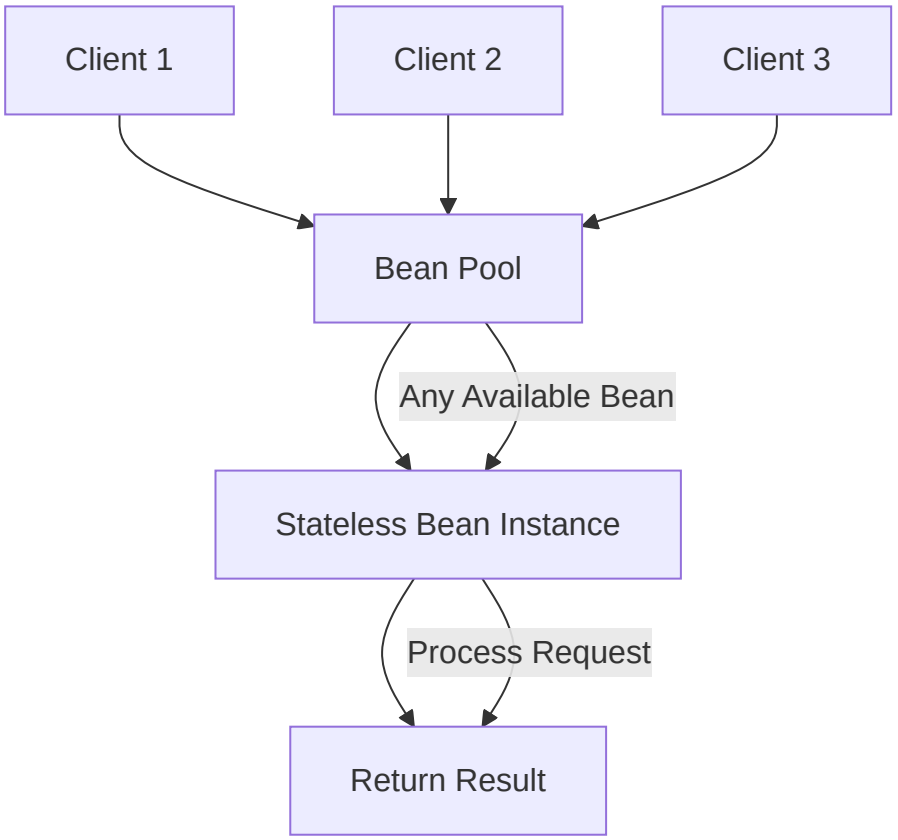
Key Definitions

> ***Stateless Session Bean:*** *A session bean that doesn't maintain conversational state with the client. Provides better scalability.*

Characteristics

- ✅ No conversational state between method calls
- ✅ Any client can use any instance from the pool
- ✅ Better scalability (fewer beans needed)
- ✅ Can implement web services
- ✅ Stateless beans are pooled by the container

Visual: Stateless Session Bean Pool



When to Use Stateless Session Beans

- When the bean's state has no data for a particular client
- When the bean performs generic tasks for all clients
- When the bean implements a web service

Code Example

```java
import javax.ejb.Stateless;

@Stateless
public class CalculatorBean implements Calculator {

    public int add(int a, int b) {
        return a + b;
    }

    public int subtract(int a, int b) {
        return a - b;
    }

    public int multiply(int a, int b) {
        return a * b;
    }

    public int divide(int a, int b) {
        if (b == 0) throw new ArithmeticException("Division by zero");
        return a / b;
    }
}
```

Key Takeaways

- ✅ Don't maintain client-specific state
- ✅ Container manages a pool of instances
- ✅ Good for operations that don't need memory of previous interactions
- ✅ More scalable than stateful beans

Topic 2.6: Entity Bean

Overview

An Entity Bean represents persistent business data stored in a database. It manages data that persists beyond the lifetime of a single client session and is identified by a primary key.

Key Definitions

> ***Entity Bean:*** *A server-side component representing persistent business data stored in a database.*

> ***Primary Key:*** *Unique identifier for entity bean instances.*

Entity Bean Types

1. **Bean-Managed Persistence (BMP)**
   - Developer writes database access code explicitly
   - More flexibility but more work
   - Tied to specific database
2. **Container-Managed Persistence (CMP)**
   - Container handles database operations automatically
   - Developer focuses on business logic
   - Database-independent

Visual: Entity Bean with Database

Key Takeaways

- ✅ Represents persistent data in database

- ✅ Has unique primary key
- ✅ Survives server crashes
- ✅ Can be shared by multiple clients
- ⚠️ Now superseded by Java Persistence API (JPA)

Topic 2.7: Servlet Overview and Architecture

Overview

Servlets are Java programs that run on a web server and handle client requests dynamically. They provide a server-side component model for building web applications.

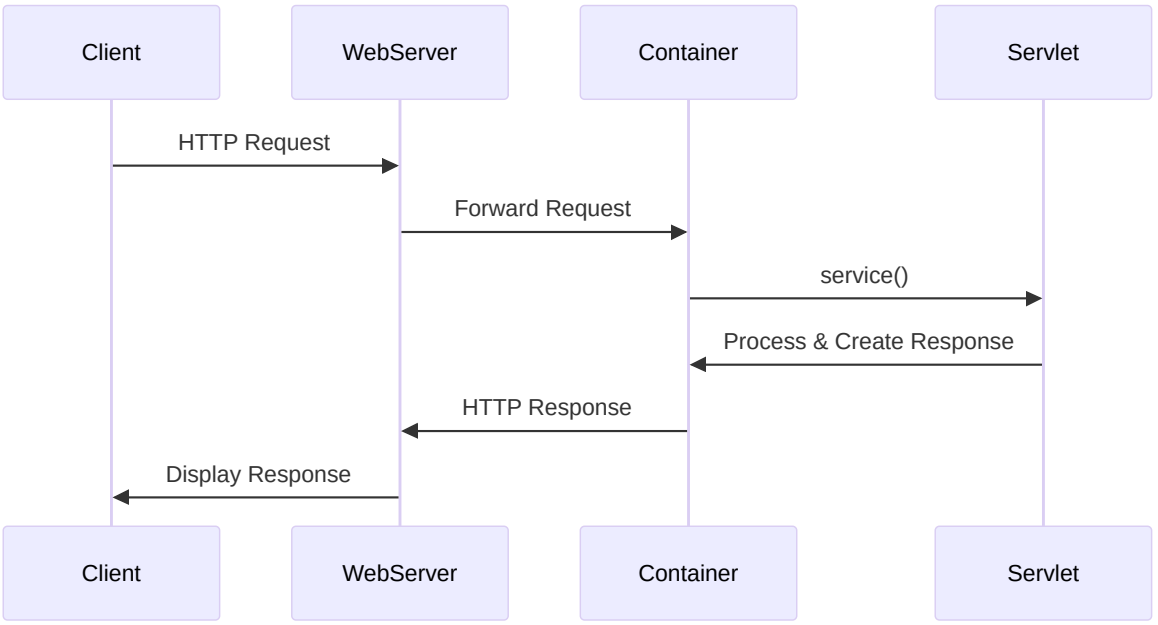
Key Definitions

> ***Servlet:*** *A Java program that runs on a web server, handling HTTP requests and generating dynamic responses.*

Servlet Architecture Components

1. **Client (Web Browser):** Sends HTTP requests
2. **Web Server:** Receives requests
3. **Web Container (Servlet Container):** Manages servlet lifecycle, URL mapping
4. **Servlet:** Processes requests

Visual: Servlet Request Flow



Servlet Advantages over CGI

- ✅ Better performance (threads vs processes)
- ✅ Platform-independent
- ✅ Robust (automatic memory management)
- ✅ Secure (sandbox model)
- ✅ Portability

Types of Servlets

1. **GenericServlet:** Protocol-independent
2. **HttpServlet:** HTTP-specific (most commonly used)

Key Takeaways

- ✅ Servlets handle HTTP requests/responses
- ✅ Run in web container (Tomcat, Glassfish, etc.)
- ✅ More efficient than traditional CGI
- ✅ Thread-based (not process-based)

Topic 2.8: Interface Servlet and the Servlet Life Cycle

Overview

The Servlet interface defines the contract that all servlets must implement. It provides three lifecycle methods and two non-lifecycle methods.

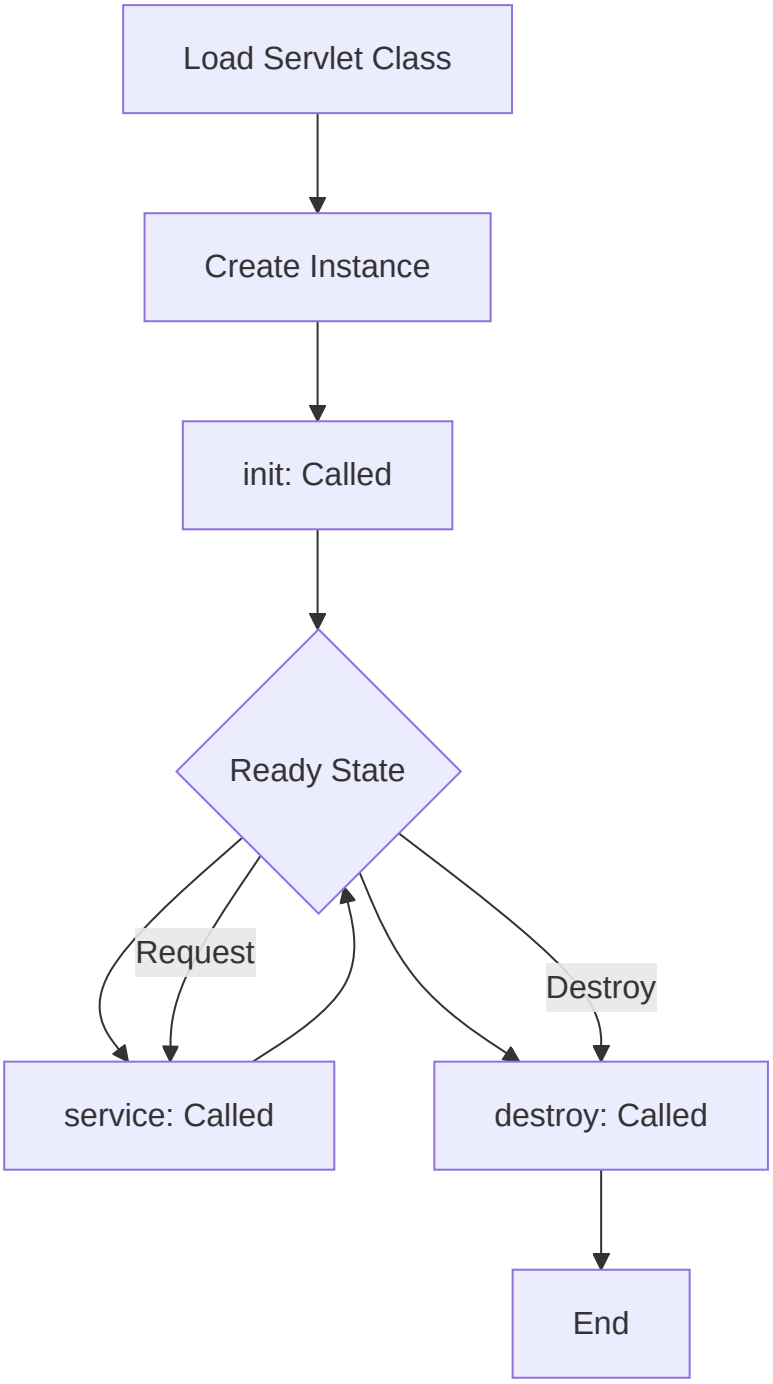
Servlet Interface Methods

Method	Description
init(ServletConfig)	Called once at initialization
service(ServletRequest, ServletResponse)	Called for each request
destroy()	Called when servlet is removed
getServletConfig()	Returns configuration object
getServletInfo()	Returns servlet information

Servlet Life Cycle

1. **Servlet class loaded** - ClassLoader loads the servlet class
2. **Servlet instance created** - Container creates instance
3. **init() called** - Initialization (once)
4. **service() called** - Handle requests (multiple times)
5. **destroy() called** - Cleanup before removal

Visual: Servlet Lifecycle



Creating Servlets - Three Ways

1. **Implement Servlet Interface**
2. **Extend GenericServlet**
3. **Extend HttpServlet** (Most common)

Code Example: HttpServlet

```java
import javax.servlet.http.*;
import java.io.*;

public class DemoServlet extends HttpServlet {
    public void doGet(HttpServletRequest req, HttpServletResponse res)
            throws ServletException, IOException {
        res.setContentType("text/html");
        PrintWriter out = res.getWriter();
        out.println("<html><body>");
        out.println("Welcome to servlet");
        out.println("</body></html>");
    }

    public void doPost(HttpServletRequest req, HttpServletResponse res)
            throws ServletException, IOException {
        doGet(req, res);
    }
}
```

Key Takeaways

- ✅ Override doGet() and/or doPost() in HttpServlet
- ✅ init() called once; service() for each request
- ✅ destroy() called when servlet is unloaded
- ✅ web.xml maps URLs to servlets

Topic 2.9: Handling HTTP GET Requests

Overview

HTTP GET requests are used to request data from a specified resource. GET is the default HTTP method.

GET vs POST

GET	POST
Data in URL	Data in request body
Limited data size	Unlimited data size
Can be bookmarked	Cannot be bookmarked
Less secure	More secure
Default method	Used for sensitive data

Code Example

```java
protected void doGet(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {

    // Get parameters from URL
    String username = request.getParameter("username");
    String[] items = request.getParameterValues("items");

    // Set content type
    response.setContentType("text/html");

    // Get PrintWriter
    PrintWriter out = response.getWriter();

    // Write response
    out.println("<html><body>");
    out.println("<h1>GET Request Received</h1>");
    out.println("<p>Username: " + username + "</p>");
    out.println("</body></html>");
}
```

Key Takeaways

- ✅ GET parameters visible in URL
- ✅ Used for retrieving data
- ✅ Can be cached and bookmarked
- ✅ Limited data size due to URL length

Topic 2.10: Handling HTTP POST Requests

Overview

HTTP POST requests are used to send data to be processed to a specified resource. POST is commonly used for submitting form data.

Code Example

```
protected void doPost(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {

    // Set character encoding for proper handling of international characters
    request.setCharacterEncoding("UTF-8");

    // Get parameters from request body
    String username = request.getParameter("username");
    String password = request.getParameter("password");
    String email = request.getParameter("email");

    // Process data (e.g., save to database)
    // ...

    // Set content type and redirect
    response.setContentType("text/html");
    PrintWriter out = response.getWriter();
    out.println("<html><body>");
    out.println("<h1>POST Request Received</h1>");
    out.println("<p>Thank you, " + username + "!</p>");
    out.println("</body></html>");
}
```

Key Takeaways

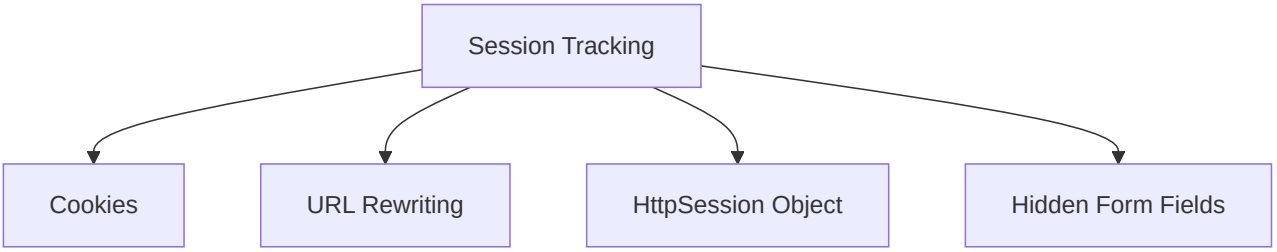
- ✅ POST data hidden from URL
- ✅ Used for submitting sensitive data
- ✅ No size limitations
- ✅ Cannot be bookmarked or cached

Topic 2.11: Session Tracking

Overview

HTTP is stateless; session tracking maintains user state across multiple requests. There are several techniques for session tracking.

Methods



Method	Pros	Cons
Cookies	Simple, persistent	Can be disabled
URL Rewriting	Always works	Breaks bookmarking
HttpSession	Server-managed, secure	Server memory
Hidden Fields	Works without cookies	Extra HTML

HttpSession Example

```
// Creating/Getting session
HttpSession session = request.getSession();

// Setting attributes
session.setAttribute("user", "John");
session.setAttribute("cart", cart);

// Getting attributes
String user = (String) session.getAttribute("user");
List cart = (List) session.getAttribute("cart");

// Removing attributes
session.removeAttribute("user");

// Invalidating session
session.invalidate();
```

Key Takeaways

- ✅ HTTP is stateless - need explicit session tracking
- ✅ HttpSession is the most common approach
- ✅ Sessions have configurable timeout
- ✅ Session data stored on server

Topic 2.12: Cookies

Overview

Cookies are key-value pairs stored on the client side. They are used to maintain state between requests.

Types of Cookies

Type	Description
Session Cookie	Deleted when browser closes
Persistent Cookie	Has expiration date

Code Examples

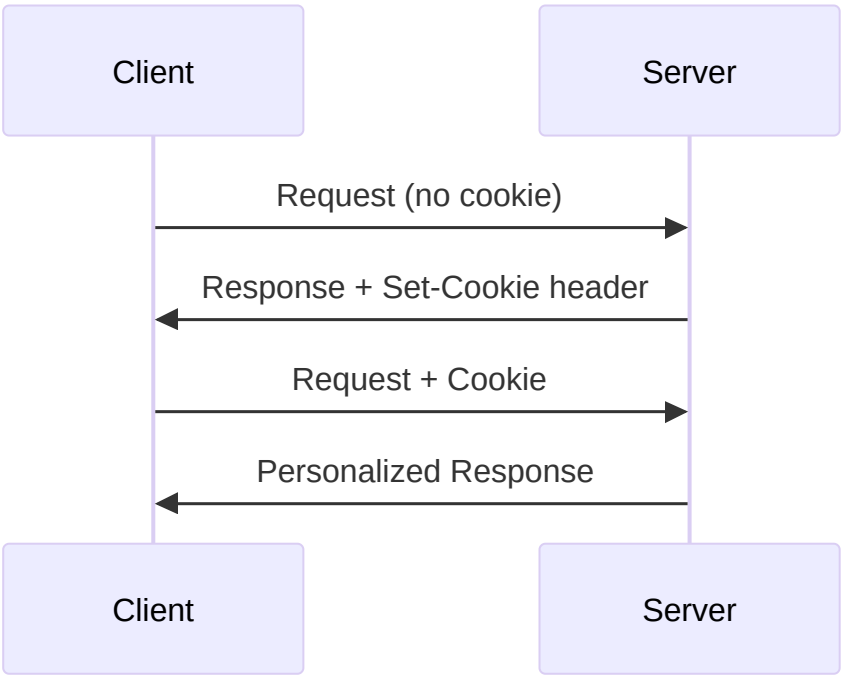
Creating a Cookie

```
// Create a cookie
Cookie cookie = new Cookie("username", "john");
cookie.setMaxAge(3600); // 1 hour (in seconds)
cookie.setPath("/"); // Available throughout the application
response.addCookie(cookie);
```

Reading Cookies

```
Cookie[] cookies = request.getCookies();
if (cookies != null) {
    for (Cookie c : cookies) {
        if (c.getName().equals("username")) {
            String value = c.getValue();
            System.out.println("Username: " + value);
        }
    }
}
```

Visual: Cookie Flow



Key Takeaways

- ✅ Cookies stored on client side
- ✅ Can be disabled by users
- ✅ Limited size (4KB per cookie)
- ✅ Sent with every request to the server

📊 Coverage Statistics

Metric	Value
Total Syllabus Topics (Unit 1 & 2)	21
Topics Covered	21 (100%)
Unit 1 Coverage	9/9 (100%)
Unit 2 Coverage	12/12 (100%)

Notes compiled for Advanced Java Unit 1 and Unit 2. Complete coverage of all topics achieved.